

[CREATE & ALTER TABLE](#)[PK & FK Constraints](#)

INSTANT

[SQL Server Module · Part 1 & 2 • Session 24](#)

DATABASE FUNDAMENTALS & DDL/DML BASICS

CREATE · ALTER · DROP · TRUNCATE · Data Types · Constraints · INSERT ·
UPDATE · DELETE · Transactions

2 Parts

SQL Server Module
DDL & DML Fundamentals

28 Slides

Interactive Presentation
Queries & Exercises

8 Tables

Instant Academy DB
Production SQL Queries

[INSERT / UPDATE / DELETE](#)[Transactions & Safety](#)



01 • DDL — DEFINING YOUR DATABASE

DDL — Defining Your Database

CREATE, ALTER, and DROP in action — building and altering database structures

— 4 MAIN SUB-LANGUAGES OF SQL

What is DDL? — Data Definition Language

SQL is divided into 4 main sub-languages: DDL (Structure), DML (Data), DQL (Queries), DCL (Permissions). Today we focus on DDL.

DDL

Data Definition Language

CREATE, ALTER, DROP, TRUNCATE

DML

Data Manipulation Language

INSERT, UPDATE, DELETE, MERGE

DQL

Data Query Language

SELECT

DCL

Data Control Language

GRANT, REVOKE

CREATE

Build a new database object (table, database, index)

ALTER

Modify the structure of an existing object

DROP

Permanently delete a database object

TRUNCATE

Remove all rows from a table (keeps structure)

— STEP 1 & 2 · CREATE DATABASE & TABLE

DDL in Action — CREATE DATABASE & TABLE

Building a new database and creating the Students table with identity primary key, constraints, and default values.

Step 1: Create the database

```
CREATE DATABASE InstantAcademy;
USE InstantAcademy; -- Switch to our new database
```



Step 2: Create the Students table

```
CREATE TABLE Students (
    StudentID INT PRIMARY KEY IDENTITY(1,1),
    StudentName NVARCHAR(100) NOT NULL,
    Email VARCHAR(150) UNIQUE,
    City NVARCHAR(80),
    EnrollDate DATE DEFAULT GETDATE(),
    Status VARCHAR(50) DEFAULT 'Active'
);
```



— CREATING ENROLLMENTS TABLE WITH FK

DDL — CREATE TABLE with Foreign Key

Creating the Enrollments table to link Students and Courses using Foreign Key constraints.

```
CREATE TABLE Enrollments (
    EnrollmentID INT PRIMARY KEY IDENTITY(1,1),
    StudentID    INT NOT NULL,
    CourseID     INT NOT NULL,
    EnrollDate   DATE DEFAULT GETDATE(),
    EnrollStatus VARCHAR(50) DEFAULT 'Active',

    CONSTRAINT fk_student FOREIGN KEY (StudentID)
        REFERENCES Students(StudentID),
    CONSTRAINT fk_course FOREIGN KEY (CourseID)
        REFERENCES Courses(CourseID)
);
```



⚠ If you try to enroll a StudentID that doesn't exist — SQL Server will reject it! IDENTITY(1,1) auto-increments the ID starting from 1.

— ADD, ALTER COLUMN & DROP COLUMN

DDL — ALTER TABLE

ALTER lets you modify a table structure after creation — without losing existing data.

Add a new column

```
ALTER TABLE Students ADD Phone VARCHAR(20) NULL;
```

Modify a column's data type

```
ALTER TABLE Students ALTER COLUMN StudentName VARCHAR(200);
```

Drop a column

```
ALTER TABLE Students DROP COLUMN Phone;
```

⚠ Dropping a column permanently removes it AND all its data. Always backup first!

— DROP TABLE VS TRUNCATE TABLE

DDL — DROP and TRUNCATE

DROP deletes table structure AND data forever · TRUNCATE removes all rows but keeps structure.

✗ DROP TABLE

```
DROP TABLE Students;
```

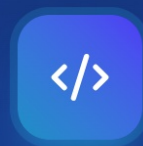
- Permanently deletes table AND all its data
- Table structure is gone — cannot be undone
- Cannot drop table with child FK references
- Drop child tables (Enrollments) before parent

↺ TRUNCATE TABLE

```
TRUNCATE TABLE Leads;
```

- Removes ALL rows but keeps table structure
- Much faster than DELETE for large tables
- Resets IDENTITY counter back to seed value
- Cannot be rolled back — use with caution

Quick Comparison: **DROP** = Table gone forever · **TRUNCATE** = Empty table, structure survives · **DELETE** = Remove specific rows (DML)



02 • DATA TYPES & CONSTRAINTS IN SQL SERVER

Data Types & Constraints in SQL Server

Choosing the right type and enforcing business rules for every column

— NUMERIC, TEXT & DATE TYPES

SQL Server Data Types for Analysts

Numeric (INT, DECIMAL, FLOAT, BIT), Text (VARCHAR, NVARCHAR, CHAR), Date & Time (DATE, DATETIME, GETDATE()).

Numeric

INT

Whole numbers. IDs, counts. Range ± 2 billion.

DECIMAL(p, s)

Exact decimals for money: DECIMAL(10,2) = 99999999.99

BIT

True/False (1/0). IsActive, IsPaid.

Text

VARCHAR(n)

Variable-length text. Most common for English names, email.

NVARCHAR(n)

Unicode text — use for Arabic names & special characters.

CHAR(n)

Fixed-length. Country codes (e.g. 'EG'), NationalID (14).

Date & Time

DATE

Date only: 2024-03-15. Enrollment dates, birthdays.

DATETIME

Date + time: 2024-03-15 09:30:00. Login logs, payments.

GETDATE()

Built-in function returning current system date and time.

Academy tip: StudentName uses NVARCHAR (supports Arabic names) · Amount in Payments uses DECIMAL (10, 2) · EnrollDate uses DATE

— PRIMARY KEY, UNIQUE, CHECK & DEFAULT

SQL Server Constraints for Analysts

Integrity Rules (PK, FK, UNIQUE, NOT NULL), Value Rules (CHECK, DEFAULT), and DDL Syntax.

Integrity Rules

PRIMARY KEY: Unique ID, no NULL

UNIQUE: No duplicates, NULL allowed once

NOT NULL: Cannot be empty

FOREIGN KEY: Links to PK in another table

Value Rules

CHECK: `CHECK(Age >= 18)`

DEFAULT: `DEFAULT GETDATE()`

CHECK (range): `Status IN ('Active', 'Pending')`

CHECK (format): `Email LIKE '%@%.%'`

DDL Syntax

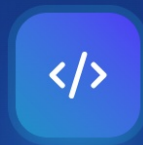
Inline: `StudentID INT PRIMARY KEY`

Named: `CONSTRAINT pk_student PRIMARY KEY`

ALTER ADD: `ADD CONSTRAINT chk_age CHECK...`

ALTER DROP: `DROP CONSTRAINT chk_age`

💡 Academy tip: StudentID uses PRIMARY KEY · Email uses UNIQUE + NOT NULL · Payments.Amount uses CHECK (Amount > 0)



03 • HANDS-ON DDL EXERCISES

Hands-On Exercises

Practice makes permanent — let's write some DDL queries together!

— CREATE THE COURSES TABLE

Exercise 1 — Create the Courses Table

Write the CREATE TABLE statement for the Courses table with all listed columns, data types, and constraints.

Column Name	Data Type	Constraint	Description
CourseID	INT	Primary Key, Auto-increment	Unique identifier for each course
CourseName	NVARCHAR(200)	NOT NULL	Full course name (supports Arabic)
Category	VARCHAR(100)	NULL	e.g. Data Analysis, Programming
Price	DECIMAL(10,2)	NOT NULL, DEFAULT 0	Course price in EGP
DurationHrs	INT	NOT NULL	Duration in hours
StartDate	DATE	NULL	Scheduled start date

[Hide Solution SQL](#)

```
CREATE TABLE Courses (
    CourseID INT PRIMARY KEY IDENTITY(1,1),
    CourseName NVARCHAR(200) NOT NULL,
    Category VARCHAR(100) NULL,
    Price DECIMAL(10,2) NOT NULL DEFAULT 0,
    DurationHrs INT NOT NULL,
    StartDate DATE NULL
)
```

SQL CODE & VERIFICATION

Exercise 1 — Solution

Complete SQL script to CREATE Courses table and verify structure using INFORMATION_SCHEMA.COLUMNS.

Column Name	Data Type	Constraint	Description
CourseID	INT	Primary Key, Auto-increment	Unique identifier for each course
CourseName	NVARCHAR(200)	NOT NULL	Full course name (supports Arabic)
Category	VARCHAR(100)	NULL	e.g. Data Analysis, Programming
Price	DECIMAL(10,2)	NOT NULL, DEFAULT 0	Course price in EGP
DurationHrs	INT	NOT NULL	Duration in hours
StartDate	DATE	NULL	Scheduled start date

</> Hide Solution SQL

```

CREATE TABLE Courses (
    CourseID    INT PRIMARY KEY IDENTITY(1,1),
    CourseName  NVARCHAR(200) NOT NULL,
    Category    VARCHAR(100) NULL,
    Price       DECIMAL(10,2) NOT NULL DEFAULT 0,
    DurationHrs INT NOT NULL,
    StartDate   DATE NULL

```

— ALTER TABLE & SCHEMA EXPLORATION

Exercise 2 — ALTER & Schema Exploration

2A: Add NationalID to Students | 2B: Modify CourseName length | 2C: Query ALL tables in database.

2A Add NationalID column to Students

Add NationalID of type CHAR(14) allowing NULL values.

```
ALTER TABLE Students ADD NationalID CHAR(14) NULL;
```

2B Modify a column in Courses

Expand CourseName to NVARCHAR(300) NOT NULL.

```
ALTER TABLE Courses ALTER COLUMN CourseName NVARCHAR(300) NOT NULL;
```

2C Explore the schema

List ALL tables in the InstantAcademy database.

```
SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE';
```



04 • HOMEWORK ASSIGNMENT

Homework Assignment

Reinforce your learning before Session 2 — build the full Instant Academy Database

— FULL ACADEMY DATABASE BUILD

Session 1 Homework — Academy Database Build

Task 1: Create Full Database (20 pts) | Task 2: Modify Schema (15 pts) | Task 3: Egyptian Business Reflection (15 pts)

Task 1: Create the full academy database

Write CREATE TABLE for Leads, Campaigns, SalesAgents, and Payments with PKs/FKs.

20 pts

Task 2: Modify the schema

Add IsActive BIT to SalesAgents & Notes NVARCHAR(20) to Attendance. Verify with query.

15 pts

Task 3: Research & reflection

Find 1 Egyptian company using SQL Server (bank/telecom) and explain 2 matching tables.

15 pts

WHAT YOU LEARNED TODAY & SESSION 2 PREVIEW

Session 1 Summary & What's Next

What You Learned Today (DDL, Schema, PK/FK) vs Session 2 Preview (DML: INSERT, UPDATE, DELETE & SELECT).

✓ WHAT YOU LEARNED TODAY

- What a database is and why companies need them
- How databases differ from Excel spreadsheets
- Tables, rows, columns, and data types
- Primary keys and foreign keys explained
- The Instant Academy 8-table schema
- SQL Server and SSMS setup
- DDL: CREATE, ALTER, DROP, TRUNCATE
- Hands-on: built Students, Courses, Enrollments tables

⚡ SESSION 2 PREVIEW: DML & QUERIES

Manipulating Your Academy Data:

- INSERT data into all 8 tables
- UPDATE student records
- DELETE leads and test data
- Bulk data operations

8
Tables

3
DDL Cmds

12+
SQL Stmts

100%
Practice



05 · DML BASICS AND BASIC SELECT QUERIES

DML Basics & Basic SELECT Queries

INSERT · UPDATE · DELETE · SELECT — Writing & Reading Data Like a Pro in SQL Server

— 4 CORE PARTS OF DML & QUERIES

Session Overview — DML & SELECT

Part 1: DML Foundations (45m) | Part 2: INSERT & UPDATE (60m) | Part 3: DELETE & SELECT (60m) | Q&A

1 DML Foundations

Understand what DML is and how it differs from DDL. Learn when and why analysts manipulate data directly.

2 INSERT Data

Master inserting single and multiple rows into academy tables like Students, Enrollments, and Payments.

3 UPDATE & DELETE

Safely modify existing records and remove data using WHERE clauses to target the right rows.

4 Basic SELECT

Read data with SELECT, filter with WHERE, sort with ORDER BY, and limit with TOP.

Part 1: DML Concepts (45m)

Part 2: INSERT & UPDATE (60m)

Part 3: DELETE & SELECT (60m)

Q&A (15m)

THE 8 ACADEMY TABLES WE WILL MANIPULATE

Quick Recap: The Academy Dataset

Our business case — Students, Courses, Enrollments, Payments, Leads, Campaigns, Attendance, SalesAgents tables.

Students

🔑 StudentID

StudentName, Email, City, EnrollDate,
Status, NationalID

Courses

🔑 CourseID

CourseName, Category, Price, DurationHrs,
StartDate

Enrollments

🔑 EnrollmentID

StudentID, CourseID, EnrollDate,
EnrollStatus

Payments

🔑 PaymentID

EnrollmentID, Amount, PaymentDate,
Method, Status

Leads

🔑 LeadID

LeadName, Source, AgentID, CampaignID,
Status

Campaigns

🔑 CampaignID

AgentID, CampaignName, Channel, Budget,
IsActive

Attendance

🔑 AttendanceID

EnrollmentID, AttendStatus, Notes

SalesAgents

🔑 AgentID

AgentName, Email, Region, MonthlyTarget,
IsActive

DATA MANIPULATION VS DEFINITION

What is DML? — Data Manipulation Language

DDL affects schema (structure) · DML affects rows (content: INSERT, UPDATE, DELETE). Can be rolled back in transactions.

DDL — DATA DEFINITION LANGUAGE

- CREATE TABLE — Build new tables
- ALTER TABLE — Modify structure
- DROP TABLE — Delete entire tables

Affects schema (structure)

Cannot be rolled back easily

DML — DATA MANIPULATION LANGUAGE

- INSERT — Add new rows to a table
- UPDATE — Change existing row values
- DELETE — Remove specific rows

Affects data (content)

Can be wrapped in transactions

💡 As a Data Analyst, you'll mainly use DML to load sample data, fix data quality issues, and test queries.

— SINGLE & MULTIPLE ROW INSERTS

INSERT Statement — Syntax

Adding new rows of data into a table using single and multiple row VALUES clauses.

Single Row Insert

```
INSERT INTO Students (StudentName, Email, City)
VALUES ('Ali Hassan', 'ali@mail.com', 'Cairo');
```

Multiple Row Insert

```
INSERT INTO Students (StudentName, Email, City)
VALUES
('Sara Nour', 'sara@m.com', 'Alex'),
('Omar Said', 'omar@m.com', 'Giza');
```

⚠ **Common Pitfalls:** Always match column count with VALUES count · Use single quotes for text & dates · Never insert into IDENTITY (auto) columns

— COURSES, ENROLLMENTS & PAYMENTS

INSERT in Action — Academy Examples

Populating academy tables + Common Pitfalls (match column count, single quotes for text/dates, never insert IDENTITY).

Single Row Insert

```
INSERT INTO Students (StudentName, Email, City)
VALUES ('Ali Hassan', 'ali@mail.com', 'Cairo');
```

Multiple Row Insert

```
INSERT INTO Students (StudentName, Email, City)
VALUES
('Sara Nour', 'sara@m.com', 'Alex'),
('Omar Said', 'omar@m.com', 'Giza');
```

⚠️ **Common Pitfalls:** Always match column count with VALUES count · Use single quotes for text & dates · Never insert into IDENTITY (auto) columns

— MODIFYING EXISTING DATA WITH WHERE

UPDATE Statement

Modifying student email, enrollment status, or agent target. 🚨 DANGER: UPDATE without WHERE updates ALL rows!

```
UPDATE TableName SET Column1 = NewValue1 WHERE Condition;
```

Update student email

```
UPDATE Students  
SET Email = 'ali.new@mail.com'  
WHERE StudentID = 101;
```

Change enrollment status

```
UPDATE Enrollments  
SET EnrollStatus = 'Completed'  
WHERE EnrollmentID = 1001;
```

Increase agent target

```
UPDATE SalesAgents  
SET MonthlyTarget = MonthlyTarget * 1.10  
WHERE Region = 'Cairo';
```

⚠️ 🚨 DANGER — UPDATE without WHERE updates ALL rows in the table! Run SELECT first to verify.

— DELETE VS TRUNCATE COMPARISON

DELETE Statement

Removing rows with extreme caution · DELETE vs TRUNCATE (condition filtering, transaction rollback, IDENTITY reset).

```
DELETE FROM TableName WHERE Condition; -- Always use WHERE!
```

Delete a specific lead

```
DELETE FROM Leads WHERE LeadID = 55 AND Status = 'Junk';
```

Delete old campaigns

```
DELETE FROM Campaigns WHERE StartDate < '2022-01-01';
```

Feature	DELETE	TRUNCATE
Removes rows by condition?	✔ Yes (WHERE clause)	✘ No — removes ALL rows
Can be rolled back?	✔ Yes (in a transaction)	⚠ Limited
Resets IDENTITY counter?	✘ No	✔ Yes

BEGIN TRAN, COMMIT & ROLLBACK

DML Safety — Transactions

How to undo a mistake before it becomes permanent: BEGIN TRAN → Execute DML → Inspect → COMMIT or ROLLBACK TRAN.

A transaction wraps multiple DML statements into a single atomic unit — either ALL succeed or ALL are rolled back.

```
-- Step 1: Start the transaction
BEGIN TRAN;

-- Step 2: Make changes
UPDATE Students SET City = 'Alex' WHERE StudentID = 101;

-- Step 3a: Check results — if correct:
COMMIT TRAN;

-- Step 3b: If something went wrong — undo:
ROLLBACK TRAN;
```

▶ BEGIN TRAN

Starts transaction block. Changes are staged, not permanent.

✓ COMMIT

Saves all staged changes permanently to database.

▶ ROLLBACK

Cancels all staged changes — returns data to previous state.

— 70 TOTAL PRACTICE POINTS

Exercise 1 — DML Practice

Apply INSERT (3 students, 1 course), UPDATE (EnrollStatus), DELETE (Junk leads with transaction rollback), & BONUS calculation.

01. INSERT 3 Students (15 pts)

Include StudentName, Email, City (Cairo, Alex, Giza), EnrollDate.

02. INSERT Course (10 pts)

'Excel for Analysts', 'Business' category, 2000 EGP, 20 hrs.

03. UPDATE Status (15 pts)

Update EnrollStatus of Enrollment #1003 from 'Pending' to 'Active'.

04. DELETE Junk Leads (20 pts)

DELETE leads with Status = 'Junk' wrapped in transaction + ROLLBACK.

[</> Show Solution SQL](#)

INSTANT

✔ SESSION 24 COMPLETED SUCCESSFULLY

THANK YOU

Thank you for your active participation!

🎉 **Session 24 Complete!** You have mastered SQL Server DDL Statements, Constraints & DML Queries!

↺ Restart Presentation

Open Session 25: Basic SELECT Queries, Filtering (WHERE) & Sorting (ORDER BY) →